

GUETTINGEN, Switzerland

WMO: 066210

Lat: 47.6017N

Lon: 9.2794E

Elev: 1444

StdP: 13.95

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 18.2	(c) 21.5	(d) 8.4	(e) 8.9	(f) 18.8	(g) 15.1	(h) 12.5	(i) 23.8	(j) 22.7	(k) 45.6	(l) 19.9	(m) 44.2	(n) 6.9	(o) 50	(p) 0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	85.7	68.7	82.4	67.9	79.3	66.6	70.9	81.8	69.3	79.3	67.8	76.9	4.5	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.4	106.4	75.6	65.8	100.6	73.9	64.5	95.9	72.0	35.6	82.2	34.2	79.7	32.9	77.1	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	14.6	12.3	14.2	91.0	4.9	2.3	10.7	92.7	7.8	94.1	5.0	95.4	1.5	97.1
			13.0	74.1	4.8	1.6	9.5	75.3	6.7	76.2	4.0	77.1	0.6	78.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	34.1	35.0	42.5	50.3	57.2	64.6	67.3	66.4	59.0	51.0	41.7	35.4
	DBStd	13.59	7.29	7.37	6.98	6.71	6.54	6.44	5.44	5.48	5.62	6.20	6.10	6.52
	HDD50	2026	493	420	245	79	13	1	0	0	3	64	255	453
	HDD65	5608	957	839	698	442	254	86	40	50	193	433	698	918
	CDD50	2194	2	0	11	87	235	439	535	507	272	97	8	1
	CDD65	300	0	0	0	0	12	74	110	92	12	0	0	0
	CDH74	2238	0	0	0	12	120	593	843	613	57	0	0	0
	CDH80	564	0	0	0	2	15	150	237	158	2	0	0	0
(14) Wind	WSAvg	5.0	5.6	5.6	5.6	4.9	5.1	4.9	4.7	4.4	4.4	4.3	4.9	5.5
(15) Precipitation	PrecAvg	42.6	2.2	2.4	2.6	3.2	4.5	5.2	4.9	4.6	3.5	3.0	3.3	3.0
	PrecMax	52.5	5.4	5.4	7.4	6.3	9.4	8.2	10.5	7.8	6.8	5.3	6.7	6.1
	PrecMin	31.5	0.2	0.7	0.9	1.0	0.9	1.9	2.1	1.5	1.7	0.4	0.4	0.2
	PrecStd	5.0	1.3	1.2	1.3	1.4	2.0	1.7	1.9	1.7	1.4	1.3	1.5	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.6	56.7	65.3	75.4	81.9	88.9	90.3	89.5	79.1	69.6	61.5	54.0
		MCWB	49.0	47.5	51.9	58.3	65.0	69.7	69.9	69.9	66.5	59.9	53.7	48.8
	2%	DB	51.6	51.9	60.5	70.9	77.3	84.3	85.8	84.2	75.4	65.5	56.5	50.3
		MCWB	46.7	45.3	49.6	55.6	62.4	68.3	68.6	69.1	65.0	58.9	51.9	47.0
	5%	DB	47.8	48.5	56.9	67.1	73.2	80.9	82.3	80.4	71.6	62.3	52.8	47.2
		MCWB	44.2	43.4	48.3	54.4	60.2	67.0	67.9	68.0	63.6	57.6	49.7	44.4
	10%	DB	44.3	45.1	53.4	62.9	69.4	77.1	78.7	76.9	68.1	59.5	50.3	44.5
		MCWB	41.6	41.1	46.7	52.5	58.6	65.4	66.8	66.6	61.7	56.1	47.8	42.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.7	49.3	53.3	59.9	67.2	72.3	73.3	72.7	68.2	62.7	54.4	50.6
		MCDB	54.7	55.3	61.7	72.2	78.3	84.7	86.1	83.5	76.1	66.9	59.0	52.0
	2%	WB	47.3	46.1	50.9	57.2	64.2	69.9	70.8	70.8	66.2	60.3	52.3	47.5
		MCDB	50.6	51.0	58.1	68.6	73.8	81.3	81.6	81.2	73.0	63.8	55.7	50.0
	5%	WB	44.3	43.8	49.1	55.1	61.9	68.0	69.1	69.1	64.5	58.3	50.3	44.9
		MCDB	47.3	48.0	55.4	64.8	70.4	78.2	79.2	78.2	70.2	61.4	52.7	46.9
	10%	WB	41.8	41.4	47.3	53.1	59.7	66.3	67.5	67.4	62.5	56.5	48.2	42.7
		MCDB	44.2	44.9	53.1	61.9	67.6	75.0	76.8	75.2	67.0	59.0	50.1	44.3
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	10.4	15.3	18.3	17.5	17.8	18.2	17.2	15.1	12.1	9.0	7.7
		MCDBR	13.1	16.5	21.9	26.0	24.6	24.3	24.4	22.9	19.4	15.8	14.0	11.3
	5% WB	MCWBR	9.5	10.7	12.6	13.1	11.6	10.6	10.4	10.1	10.2	10.1	9.6	8.7
		MCWBR	12.1	15.7	19.8	23.6	22.1	21.9	22.4	21.3	18.2	14.9	13.4	10.6
(40) Clear-Sky Solar Irradiance	taub	taub	0.292	0.317	0.354	0.391	0.404	0.411	0.412	0.405	0.378	0.348	0.317	0.288
		taud	2.436	2.368	2.342	2.272	2.277	2.298	2.287	2.325	2.380	2.450	2.484	2.446
	Ebn at Noon	Ebn at Noon	252	268	274	273	273	271	269	266	262	253	240	239
		Edh at Noon	21	28	33	39	41	40	40	37	32	26	20	18
(44) All-Sky Solar Radiation	RadAvg	355	617	1000	1368	1586	1818	1748	1527	1141	713	393	290	46
	RadStd	39	86	134	195	161	167	159	157	118	77	47	46	46

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (110 neighbors)	N/A	N/A	N/A	N/A	+0.79	+0.67	N/A	N/A	+97	+27

Nomenclature: See separate page